

I: I wanted to start by acknowledging that teaching and learning has been significantly disrupted in the past months due to our pandemic. For any of the questions I'm about to ask, please feel free to answer with reference to your normal teaching practices rather than the teaching practices you've had to adapt to during our crisis situation.

UCSB06: Ok

I: So I first wanted to start by hearing about your experience teaching undergraduate, so can you briefly describe your experience and then how does your teaching relate to your current or past research? And what courses do you currently teach that work with data? And I can repeat any of that if you need.

UCSB06: Ok so you want to hear about all of my teaching, not just the class that has to do with data?

I: So we can start by talking if you wanted to start with that class, that's totally fine if that's the most relevant application that you have for this interview.

UCSB06: Okay well I guess just more generally speaking, I have been teaching at UCSB since [year redacted] and I taught a couple of other places before that as I was finishing my PhD and just afterwards. I am in the Department of [redacted department] and I teach regularly these days. I teach a [level and course title redacted], so that's directly about data and data analysis. At the end of graduate level, the other classes that I teach have to do with [course topic redacted]. And at the graduate level, I teach again [course name redacted] and also [course name redacted], so those are the classes that I teach that are in my regular rotation here at UCSB. What else did you want to know?

I: So also just for the point of clarification, we're specifically in this case referencing those courses you teach for undergraduates.

UCSB06: Ok

I: So just for the sake of clarity. And then aside from those classes you've just mentioned, how does this relate to your current or past research that you do as well?

UCSB06: Well oddly, you know, a lot of my teaching doesn't line up directly with my research and that's kind of an artifact of a couple of things that's a little bit unusual in our department typically. It's more typical that faculty will teach, say a speciality course at least here or there that directly has to do with their research, but in my case, that's not really true because I do this [course name redacted], which is just generally about, you know, research in general. And then the [course class and topic] that I teach undergrads, that's a [course level redacted] class and that's more of just a passion that I have. I've done a tiny bit of research having to do with [course topic redacted], but it's not my area of focus. So, if I wanted to create some new classes, I could definitely link, you know, my teaching more directly with my research. I haven't

been, you know, eager to do a lot of new course preps, so I haven't done that. I also do interact with undergrads about my research because we do a lot of undergraduate...we have a lot of research assistantships for undergraduates in our department, it's a very common thing for our undergrads to do, so that's working in very small teams on a particular project, and those projects do have to do directly with my research. So, I guess in that sort of class, it's called the [course name redacted] - teaching and interns happen there that's directly related to my research.

I: Ok. Could you expand a little bit on what your research is?

UCSB06: Yeah sure. My primary area of research has to do with [research topic redacted]. I'm looking at the social impacts and implications that they have. More particularly within that, I have established a thoroughly long research agenda looking at the [research topic redacted], so the [research topic redacted] particularly as it's complicated in digital environments. And people used to hear me say that and go like "oh, you know, that's nice" you know with this kind of attitude like "hmm" you know, people you know they can't figure out you know what to believe in, they're just stupid right? But ever since, now when I say what I study, people are like "wow that's really important" so

I: Yes, I would imagine so.

UCSB06: I also study issues of [research topic].

I: Great, thank you. So, we're going to move into this whole getting data section. So, a little preamble. So, in your courses, how do your students usually obtain the data they engage with? Are they either generating it themselves, doing a search to identify pre-existing data, or do you provide them with data sets?

UCSB06: Ok. So now I'm going to switch to talking about my [course name redacted] class which is, this is a statistics - it's [course name redacted] for the social sciences. It's in the department of [department redacted], it serves as a [course level redacted]. And that's the only class in which I have students get, you know, like numbery data. In other classes, I might have them do literature searches, but I don't think that's what we're talking about here.

So, in the [course name redacted] class that I teach, I do provide students with data. We have a data set that we use. It's introduced in the first week of class. It's actually kind of loosely modeled off of the student evaluation survey just because that's something that most students are fairly familiar with, but it's kind of, it was created and tailored for this specific class because it has some things about....it's a hypothetical study in which students have taken a class either live or actually ironically remote through some kind of online learning type of thing, and then they either have a male or female instructor and they either have a certain, there's these different type of structures, and so we have it's a pre-generated data set that was created for this class over 20 years ago, I didn't create it myself, but it's still fairly easy for students to grasp because they know the concept of a survey and they know the concept of course evaluations

and so, in chapter 2 of our text book - this is a custom textbook that was made for again this class over 20 years ago, the data set and the hypothetical study are introduced, and then the students work with that data set every week so you know when we might be talking about t-test, another week we might be doing bivariate regression, and we use the same set of variables in that data set to demonstrate and illustrate those various statistical techniques that we are talking about.

So, they have, the sections for coming to section are two hours long. The first hour, the students meet with their TA in a classroom and it's been, you know, answering questions, going over their homework, those kinds of things. And then the second hour, they actually moved to a computer lab on campus - obviously before COVID. And the TA leads them through a set of computer exercises that are data analysis exercises that comes straight out of this custom textbook for our class. And so, each chapter focuses on a different thing, we start out with just descriptive statistics and then we show them how to, you know, calculate frequencies and percentages and means and standard deviations and draw histograms and look for outliers and stuff like that. And then as the class progresses, we move into hypothesis testing and we do have them run T-tests and ANOVAs and do a little bit of regression. We keep it pretty simple - don't get into you know you can - I don't go into covariates much like it's just like the basic "this is an intro class" but that's how the data analysis part of it works. It's done in the computer lab and then the students are assessed, not based on whether they themselves, you know, can figure out the right buttons to push in a data analysis software package, but rather we give them outputs that are similar to what they've seen in the lab and ask them like, you know we give them a scenario that a researcher wanted to know if "blah blah blah." Below you'll find - we use SPSS - so below you'll find the SPSS output for this study and then basically the question is they need to interpret the meaning of the output, they need to understand what the numbers tell us about what the original research question or questions were.

I: I see. So, I think I'm going to pull back out just a little bit to ask about what's the process that students actually go through to get access to the data set you have for these evaluations. Is it something that you just supply in the learning management system or is there a different means they get to it?

UCSB06: It used to be in the old days, we had the computer labs that we were using preloaded and so there was just a folder on the lab computers that the students would open up each week. Because of Gauchospace, I just make the data set available on our Gauchospace course page now and so every week when they go into the lab they get into the routine they just open it up. And again, we use SPSS. I'm thinking about maybe changing that to R but I have to learn R to do that, and I haven't invested the time to do that yet. So far we've used SPSS in the class, and I kind of sell it to the students as you know, a lot of jobs out there might find it useful that you have some familiarity with a data analysis package and so you know we're going to show you how to use this, we're going to show you the basics of it, and you should put it on your resume that you have rudimentary knowledge of SPSS which is one of the most commonly used software packages in the industry. So, I think that's fairly effective, like they're like "oh wow that's cool".

I: Yeah, thank you. That helped a lot. So, it sounds like you've only had this instance in this particular class where you supply the data set, there's no other methodologies where they have to go and find their own at any point correct?

UCSB06: No, not in this class. In my department, we do have [course level redacted] thesis in which they do independent research, well they're mentored by faculty, but they come up with their own research question, they collect their own data. So it is possible for undergraduates in our department to do that. It's a year-long program.

I: Ok, so in that case, I mean is there any kind of standardization of, like, how you direct students to collect? I mean obviously, there's methodology within the research itself or your discipline, but I'm thinking is there any structured guidance that's given to them about how to go about doing that?

UCSB06: You know, in this case, and because we're different from other fields that might have kind of common data sets that people, you know, might be able to access. I mean, certainly that's a possibility for our students, but more typically, they're asking some research question where they need to generate their own data. Again, this is in the [course level redacted] program. And so, they work for a quarter, it's one quarter long where they develop - they're in a class that's led by a faculty member where they're developing hypotheses and ideas. You know, each person for their own project and also starting to think about the methodology, so sometimes the research question is going to lead to a more qualitative study maybe, it's just interviews, something like that. Other times it's going to be to a methodology with quantitative data. Typically for us, it's more like surveys and experiments. I suppose someone can do big data, but our students typically don't come in with the skill sets to do that to, you know, to crawl in and pull-down data. Although some could probably do it, but so it's wide open. So, the possibilities are kind of endless there and so the guidance that they get is one-on-one. This is a small program, so the guidance they get is pretty much one-on-one for developing their ideas both in terms of honing in on their research questions as well as the methodologies to address those questions. Then in the second quarter, they're supposed to be collecting data, and if they're able to collect it in the second quarter, start to analyze it. And if they can't collect their data in the winter, if they don't finish until the end of winter, then the data analysis will start in the spring. And then at that point, there is another faculty mentor who is available to help them do the data analysis. And I think that the reality of that and you could talk to that person like I can give you his contact information. He gets course credit for it. I think that the reality is he pretty much does the data analysis for the students because they just don't - they've taken the introductory class but the introductory class is often too rudimentary to really serve the needs of these more complex research questions that they are asking, so he really kind of leads them by the hand, sits down with them, literally sits down with them in front of the computer. He has, I think, he tells them how to clean their data sets, but once they're clean, they meet together and do the analysis together. And he points out things as they go along, but there's no reason that we can expect the students would know how to run a (...) I was just looking at a, you know,

moderated meditation analysis using you know process macro. There's no way an undergraduate would know that.

I: Hmm okay. So then this might be touching a little bit on what you've already mentioned before in the intro course, so feel free to repeat a little bit if you need to. So, we're looking here for the how they work with that data, so how does students manipulate, analyze or interpret data in your courses? So, you already mentioned a tool, SPSS. Are there other tools that students use as well?

UCSB06: Not in my class.

I: Okay. Anything for visualizations?

UCSB06: We do in the early part of that class, SPSS, we have them produce histograms, line graphs, you know, you can do box plots, you know, it has within SPSS, it has its own kind of data visualization. Nothing fancy. And the class, like I said, we keep the class very, very simple and this is not an advanced class. This is a class for students who have an absolutely terrible fear of math and a fear of quantitative data and so we keep it very simple.

I: So then, on the heels of that, I think I might know your answer. But what prior knowledge of these tools or software do you expect students to have? And this can even be like most basic of all understanding of math or...

UCSB06: Zero. Well, not zero. I tell them the first day that if, you know, most of them are worried about the class because, you know these are not engineering majors, these are not even economics majors, these are social science majors who like never dreamed they would have to take a statistics class. So, I tell them, you know, "I know you're nervous. However - and if I'm in person, I do this by raising a hand. I say, "Okay, by show of hands how many people in here can add?" And then I have them "How many people here can subtract?", "How many people here can do multiplication?", "How many people in here can do division? Not even long division and you'll get to use a calculator." And so, everybody, you know, was raising their hands and so I'm like "Okay, if you raise your hand, you for all, for those four things and remember you can use a calculator, you are well prepared to succeed in this class.

I: Okay, great. And I guess on the other side of that, this question is pretty broad. This is "what do you teach them explicitly?" And I think this is in reference to the tools that you've already kind of gone through a little bit about you know like what kinds of features you're trying to get them to engage with the tool?

UCSB06: Okay yeah, that's just, you know the tool just follows whatever, our use of tool just follows whatever concept we're studying. So for example, the first week is we introduce them to this hypothetical data set and we have them do some data entry so we're just we're just kind of opening up SPSS, we're explaining the environment like over here is where you can input data and over here is where you can look at a list of your variables and over here is you know, so we

just kind of orient them to the interface of SPSS the tool. And then we have them do a little bit of data entry just so they can get the hang of that. We have them label variables, we give them like the variables already have names that then we have them you'd have like variable labels that it's you know a longer more descriptive name for the variable so that they don't have to remember that like a variable that's abbreviated actually like "exp" actually refers to the expertness of the instructor, something like that. Then we have them do some labeling of values. So, for example, if there are categorical variables you know gender we teach them that okay in our data set one it means female and two means male. We have them enter that in so they get familiar with data entry and setting up basically the codebook to understand the data. That's week one. Week two, they start with doing just some basic descriptive statistics. You know, we have them use the tool to do the you know the mean, the min, the max, the mean, the standard deviation, you know really basic stuff and just picking out a few variables here and there that "the instructor wanted to know the average... and how students perceived the instructor on average...or you know, the average rating of the course overall," and so they'll look at the average and we'll talk about the standard deviation overall. The third week is where we do some of those graphical representations - histograms, line graphs, pie charts, again through the tool. And then after that, we start getting into hypothesis testing, but - oh Z-scores, having them calculate z-scores, show them how to do that - but a lot of my class is actually less about the computation and it's more about the understanding. Like they do, like for a T-Test, I tell them....well for variance for example, I say you know there's a formula for calculating variance and most statistics classes are going to make you learn that formula and memorize that formula. And that's great if you want to look the formula and memorize the formula, do it. I am much much more interested in you being able to tell me what is the meaning of variance? What information does this give you? Why would you want to calculate a variance? You know, what kind of question can it help you to answer? So, I'm much more conceptual than I think most stats classes are. So anyways, so then back to your question, though, about the tool, we just use whatever topic we're teaching, we use the tool to illustrate - you know, okay now we want to know if there's a difference between students' rating of the course, if they had the course online versus live? And, so we have them do a t test and then again the focus is not how do you produce that t-test, the focus becomes okay here are the results, here's a bunch of numbers on the page you know how do you make sense of these results from the SPSS? So we take them through the output, or what we call the output, and we show them how to look at the numbers you know what to look at, what they mean, and how to then translate what that statistical output means in terms of okay how can we answer the question of is there a difference between how students rated the course if they took it live versus online and then they have to kind of like translate the statistics into words. And so, we just continue doing that with each statistical technique that we cover in the class. We go up to, we do, because the conceptual nature of the class, we go up past what most [course name redacted] classes do, I actually have them do some multiple regression, again just some basic stuff, but looking at multiple predictors of one variable and asking questions about you know do all of the hypotheses predictors help to predict the dependent variable? Do only some of them do? Which ones predict better or worse? And have them answer questions like that.

I: I see. So, from what you've explained so far, it sounds like this particular tool SPSS is crucial for the teaching of this course. Is that accurate or...?

UCSB06: It is, although there are many other programs besides SPSS.

I: Right, you mentioned R...

UCSB06: Yeah R, SASS, I think there are many, many others out there. I mean you could probably do most of what we do in Excel. I don't know how to do it personally, but I can probably figure it out.

I: Mm hmm okay. So then, other than you already mentioned like there sounds like students kind of come in with a lot of reluctance or like fears about their capabilities. Are there any other challenges that you face when you're working with students to teach them about working with data?

UCSB06: I think the fear is the biggest one and just getting students to actually do their work. That's not specific to a data class, it's any class, but the students that I find who struggle the most are the ones where it turns out that they really haven't put much effort in, they haven't bothered to do the readings, they haven't bothered to turn in their homework assignments, you know that sort of thing. And then, they come like "I don't know why I did so poorly on the test" and it's like really? It's pretty obvious to me. But that's not specific to data in particular. I think the thing that's most relevant to classes involved in data analysis is just students' fear of math. And that's going to vary across the divisions, right? I mean, I'm sure engineering students have a lot less fear of math than humanities and social sciences students do.

I: Okay, so how do the ways in which you teach with data relate to the learning goals of your field more broadly?

UCSB06: Ah okay. Well in my field, which is similar to a lot of social sciences, there are kind of two sides or two approaches to research - more qualitative approaches and more quantitative approaches. Historically, my department has taken a quantitative approach to [discipline redacted] research. It's actually kind of in the process of changing right now. But this class was set up back when - or the purpose, really the reason that we set up the class that I teach in the lower division level is because a number of faculty in our department teach research - quantitative research - so they're presenting findings of studies and they actually have students read the quantitative, you know, the journal articles that are kind of presenting quantitative research and [discipline redacted]. And so, the idea was we want students to have a basic familiarity with quantitative research and data so that when they are assigned these journal articles that present the results of experiments and surveys and things like this, they don't just kind of shut down when they get to the analysis section. You know, if they're seeing a bunch of Greek letters and they're seeing P's less than .05 and they have no frame of reference for this, they often will just give up. And so, that's why we developed this class, which I actually the class that I teach, I actually kind of bill it more as more of a statistical literacy class than it is an actual

math class. It involves some math, but like I said, basic simple stuff and what we're really teaching them is kind of like "why do people do data analysis?", "what kinds of questions that's useful to answer?", and "how do we interpret the results?".

I: Okay. So you mentioned that this is specifically relating to their needs as students as they engage an academic text or literature, but do you ask them to think critically about those sources that they encounter in their everyday life as well?

UCSB06: Yeah actually, very much so. The first day of class, I kind of go through like you know a lot of you out there are probably wondering why do you have to take [course name redacted]. "I want to major in [discipline redacted] or another special science, so what good is knowing about [course name redacted]?" I do this whole song and dance about you know both what I said earlier about you know when you get into the upper division, when you get into a major or you may have to read these kinds of things, but besides that we're surrounded by statistics all the time. We see you know reports, media reports and then if you go into your job, you are going to have to be able to interpret data if not you know collect data and analyze it yourself. You know, a lot of our students go into intellectual marketing and if you want to understand your markets, you want to understand consumer perceptions, you're going to be, you know, if not doing the research yourself, you're going to be trying to interpret research that someone else has done. And so I kind of give them a number of reasons including that it's important to be a good information consumer. You don't want to just hear something and then take it in without being critical, you need to be a critical consumer to understand if you should believe what you're being told.

I: So when you reference their potential future career aspects, are there...do you ever call attention to certain - I guess - exercises where you like "This is something that you'll be commonly asked to do and your professional contexts of...."

UCSB06: Yeah, I mean not so specifically. I say you know a lot, if you want to be....like I try to work within the majors that, or sorry, the careers that [discipline redacted] majors tend to gravitate towards, [redacted professional activities related to the major], I mean it can be anything, but a lot of those kinds of careers, and so I will say like....I've done this before like, "How many of you think you might want to be a [profession related to major redacted]? You know raise your hand", and then I say, "Well in [subfield redacted], you are going to be, you know, receiving information that you have to make sense of and then translate for your audience, and a lot of that information is going to have numbers, right? It's going to be statistical information. So, I go through that. Or like I do the whole thing with you know if you're going into [areas redacted] research, you're going to be collecting data from consumers, you know, or looking at sales figures. Or even if you're [area redacted], you're going to be looking at statistics that are compiled about you know people within your organization or how your organization functions inside. I kind of go through some general examples like that, kind of hitting some of the careers that I know that our majors tend to gravitate towards.

I: Gotcha. Great, thank you. Have you observed any policies or cultural changes at your institution that influence the ways you teach with data? And you just mentioned the shift from quant to qual, but any other shifts that you've noticed as well?

UCSB06: Yeah, one thing that I have noticed and this isn't in my own teaching, but I thought it was interesting, so the class that I teach - there isn't - is not the only class that's a [course name redacted] that also fulfills [course level redacted] requirement at the university. There are two other classes that are similar, and students therefore have a choice of which one they want to take. And I just found out recently that the one that's being taught in [department name redacted] redacted - I think it's [course name redacted] or something like that - has added sort of a big data component, which I thought was really intriguing and I wanted to find out more about how they're doing that because that's something that I think could be pretty exciting to to work into my class. So I haven't done anything, COVID, you know, has just made it such that all I do is try to keep my head above water with regard to teaching, but you know, once this settles out, I think I'm going to reach out to them and ask to see, you know, maybe talk to the instructor to see what they're doing with that because there's a fine line. Again, these are students who are not strong enough, resent having to take anything related to math or data analysis and I don't want to overwhelm them, so I want to see how they, how they reached out to those kinds of students with big data. So, it's not an institutional thing, it's just you know a [department name redacted] made that choice to incorporate some of these new more computational kinds of approaches into their data analysis class. Yeah...But institutionally, I can't really think of anything that's changed.

I: So, this is kind of related, but do instructors in your field face any ethical challenges in teaching with data?

UCSB 6: ...not sure what that question is getting to. All of our data pretty much involves human subjects, so we get human subjects approval. This is for our research though. Teaching...the only thing that comes to mind with quantitative data it's anonymized right like we're not really associating identities a person identifying information with it in a teaching context, and we're very careful to avoid doing that in a research context also. The only thing I can think of is if you're talking about maybe interview data where you haven't secured permission of the interviewees to show that in class, like again, this is more on the research side. I've been involved in research projects where we have collected data in which participants were videotaped and we are very careful to ask and we get permission ahead of time for them to say yes, or no, uncomfortable, with your showing clips of my video in a classroom. So that's how we typically... I don't know. Can you give me an example of ethical issues?

I: I guess one of the things that came in my head was actually not in reference to this intro course but into the honors course, especially when students are being asked to collect or generate their own data set essentially.

UCSB06: Oh. Oh. Ok.

I: We know with your other course, you supply a dataset that is 20 years old and has been anonymized, so for me I don't see a connection....

UCSB06: ...and these are not real peoples' responses. So, for the [course level redacted] honors project they go through human subjects training and they have to get approval. We follow all ethical guidelines for conducting research and data collection.

I: ...and then for the sake of exploring it, you mentioned there was an individual who helps them with the analysis portion of that and that's the only other area I could see... maybe less-so of an ethical issue.. But how they have another individual stepping into the process and whether or not the IRB process that we've just been through for this research project as well, and how that shifts things.

UCSB06: Those are just datasets that are anonymized. There are no names associated. The faculty member who also has human subjects training in general for his own research is never seeing the names of anyone. Literally like a spreadsheet of numbers.

I: We are now into training and support. Time check, it looks like we are doing pretty well. So, in your courses does anyone other than you provide instructional support to your students in obtaining or working with data, of course you already mentioned this other individual for the honors class, but are there others outside of that individual?

UCSB06: TAs. Whoever is teaching, a TA in the class would be involved.

I: How does their instruction or support relate to the rest of the course. I mean, it sounds like, if they're doing... are they mostly leading individual sections? I know TAs kind of function in different roles depending on the course. So, could you explain that a little bit further?

UCSB06: In our courses they lead sections and so in my data, in my classes that do data analysis they are... they go over the homework problems with students and help them understand because students do data analysis by hand there and they lead the lab sessions. So, they are the ones teaching SPSS and showing them how to obtain the outputs to then interpret.

I: So, then how are you communicating with them about this instruction or the support that they are having to provide to the students in this process?

UCSB06: So, we have... so whenever I get new TAs I meet with them and I ask them about their own you know experience, comfort level, familiarity with SPSS and all of that. I give them access if they don't already have access to SPSS for example. Typically, the way that our TAs are assigned is it's a match process, so I'm typically not faced with a situation where I have a TA who has zero knowledge. So, it's, they typically come in with some kind of background and a comfort level with [course name redacted]. So that isn't too much of an issue. I have had... it varies a lot. You know, a huge range, and so I have a weekly meeting with my TAs and in that

weekly meeting we... my primary goal is to make sure that they are ready to go into those sections/lab sessions with the students. So, they they do the assignment ahead of time themselves. So, they will go through the computer problems, they will work them out. I also provide answer keys, sample output for the undergraduate students as well. They have instructor's manual and then I ask them every week, I tell him you know, "Go through the problems ahead of time before our meeting so that if you have issues you can bring them up in the meeting and we can go through it together." Because the class stays at such a simplistic level, I haven't really had to do very much. The TAs have been fairly comfortable. I haven't had a lot of issues with the TAs. Once or twice, I will have a TA come and say, "Ugh, can you just double check" or "I don't understand this..." and I meet with them maybe either privately or just in the context of our regular weekly meeting to make sure that they feel confident walking into the room.

I: I'm curious, how many sections are you working with a quarter?

UCSB06: How many TAs, or?

I: Yes... that's a good question. I was making an assumption that there was only one TA per section, but maybe that's not the case.

UCSB06: So, we have sections weekly and it depends on our enrollment. For every 75 students we get we get one TA. So right now I have a class of 75, I have one TA. But that one TA teaches three sections. So, in other words, the sections are 25 students each and the workload for the TAs is to teach three sections per week. They always occur at the end of the week, like right now. There's a section running right now that [section times]. So they are two-hour sections, so it's six hours that each TA spends with 75 students.

I: Ok. How many times is the course offered in an academic year?

UCSB06: It's offered either [range redacted] times a year. So this year it is offered [number redacted] times. I think that my department has kind of ... it used to be offered [number redacted] and now we're offering it all [number redacted] quarters.

I: Okay. So, to your knowledge are their ways in which your students are learning to work with data outside of this formal coursework?

UCSB06: I think that would just be on an individual interest basis and only if the students chose to share that with me would I have any insight into that. So, maybe.

I: But you haven't heard?

UCSB06: Nah. You know, it's not the sort of thing a lot of students would share, and by the way, our classes are huge. So, we don't really get to know our students. Sadly its one of my biggest frustrations with the university. You know, a small upper division class is 80 students. We're not

having idle chit chat a whole lot personally unless they happen to come into office hours and ask. I don't get a lot of students mentioning to me off the cuff that they're training... you know, teaching themselves data analysis.

I: Do you ever encourage, for the sake of following up on it, do you encourage them to other resources that they could reference outside of the coursework you are providing if they wanted to get further training?

UCSB06: I mean if a student did come to me, and if they were to say... actually, a lot of my grad students do that. They say, "This intro course has really got me excited about stats, how can I go learn more about this" and if an undergrad did the same thing I would tell them the same which is - there are a lot of other classes here at the university that you should look into. There is a great new data science program that the university has, I'm forgetting the actual name of it, but they have an intro course to data science and you know they can take P-stat courses, they can take classes in psych, they can take classes wherever they can find them if they want to pursue that interest. I would refer them to ... the [discipline redacted] major really doesn't offer... at the undergraduate level doesn't offer anything else. All of our stat classes besides this intro-level course, the course that I teach, are at the grad-level. I just tell them to explore their options. They can also nowadays do like Coursera or other online options as well if they want to pursue that. That is not something that our students as [discipline redacted] majors really ask a lot about.

I: Sure. So this is pivoting more to your own training now. Have you received training in teaching with data other than your graduate degree? Examples would include a workshop, a technical support program, or help from peers.

UCSB06: Yeah, here and there I have. I think that it's actually sort of tragic that we don't have more opportunities to do this cause statistics have changed since I was in grad school in [year redacted]. There are a lot of common statistics that are used today in my field that I was never trained on because they did not exist. So, I really wish that there were more opportunities. I say that... there are opportunities and I am aware of them. There are R workshops, there are... I actually have participated in... there is a summer workshop that was done through the education school on specific topics, and they're really helpful. It's just very, very hard to fit it into our busy lives. I mean we're not just teachers we're, in fact we're first and foremost researchers and that's a full time and half job by itself. Then we're teachers. That's another full time job. Then we're asked to you know administrators and we sit on all kinds of committees from the department to the campus to our field-level. So, fitting that in is very, very difficult. So, typically...usually for faculty in my field if you need to use a statistical technique or a data analysis technique that you don't know, you either teach it to yourself. You can reach out to colleagues that know, or you can look online, or you can try to participate in... the I've been trying to participate in an R workshop for 3 years now and every single time they are offered during the time that I teach. Every time. Or, I've started to do some computational work or work that requires computational analyses with big data and I don't have the skill set to do that. I

partner with computer scientists. So, I collaborate. I find people who have the data analysis skills that I don't. So that's how I do it.

I: This is a slight deviation, I think. Do you use any data sets, assignment plans, syllabi, or other instructional resources that you've received from others. Or, do you make your own resources available to others related to teaching with data to undergrads.

UCSB06: I would say I do both. Like the class... I inherited these materials when I started teaching his class from others and because I felt they worked well I kept them. If I didn't feel they worked well I would have looked for something else or created my own. And as other people have begun teaching his class if they have asked me to share my resources and datasets and all that I'm happy to do so. So, I can imagine like if I were teaching some kind of specialty course and I needed to create resources, I would do that myself. So I think it runs the range, but the easier thing is to take stuff that has already been out there and use it. But if you need to create something yourself, then we have the ability to do that.

I: This is a big picture question. Considering the evolving trends in your field that you have just referenced, what types of training or assistance would be most useful for instructors who are teaching with data?

UCSB06: I think that the workshops are good and I think that they should continue. I've just been frustrated that the timing, literally I don't know what it is, but the timing has never worked out. I do think it would be nice if there was a bigger effort on campus to... toward what I would call maybe data professional development because there are changes that happen in the data analysis world and legs we all kind of scurry around teaching ourselves. But it would be nice if there was kind of a one-stop-shop or a place that we knew that we could go get help, or that ran regular workshops that might vary the time here there - that would just provide more support for those of us who need to learn new techniques. Typically though, the need to learn new techniques is more in the are of research than it is teaching. But there is a lot of overlap between those two for people, so it could be both.

I: Final question. Is there anything else from your experience or your perspective as an instructor on the topic of teaching with data more broadly that you'd like to share?

UCSB06: For me I think that a number of students, or the type of students I teach, that is students who are not gravitating towards the physical sciences or math majors or engineering, so in other words, I'm talking more about the humanities and social science students... I think an emphasis on the conceptual understanding of data analysis and not just the procedures and not just the math... I have a graduate student right now, or there is a graduate student right now in our department who...she's [nationality redacted] and comes incredibly, amazing data analysis skills. It's incredible. Can pick up the most complex, or do the most complex analyses. But whenever I say, "Ok. Tell me what that means." It's like hitting a brick wall. We have a good relationship. We tease each other, but I'm always like, "It's great that you have these skills..." and she's always like "I want to learn more skills, more skills, more skills!" I'm like, "That's great!

Do it. But now you need to translate. Now you need to be able to take those math skills that you have"... you know looking at matrixes and complex visualizations "and now you have to be able to tell somebody...tell somebody that has not had mathematical training what the heck these pretty data pictures mean. Or what the heck these numbers mean." And she's like, "Oh well, like you can see that this number is higher than that number" and she stops there. But, what does that mean for what you are researching? So, I really think that teaching students data analysis has to not just be here are the procedures or here's the math of it and not just... you know conceptually you have to somehow marry those two things together and emphasize both equally.

I: Great! Thank you again for your time. I'm going to go ahead and stop the recording.